

Vantage Prep

SAT Math Foundations — Unit 1 Curriculum

Linear Algebra: Heart of Algebra & Linear Functions

Program	Detail
Instructors	Agastya Kondamuri / Aditya Garg
Grades	6th – 8th (Bret Harte Middle School and equivalent)
Length	2 weeks · 7 sessions (Mon / Wed / Fri each week, plus one Saturday office-hours session after Week 1)
Cohort Focus	This first cohort is 100% Linear Algebra — no quadratics — to go deep on the SAT's single most-tested skill area

1. Why Linear Algebra First

Every SAT Math test includes an Algebra domain built entirely from five official skill categories — and all five are linear-algebra topics. This isn't “extra” math on top of what Bret Harte students already do: it's the SAT's version of regular Pre-Algebra and Algebra 1 coursework, resequenced and drilled with SAT-style questions.

Official College Board skill (SAT Algebra domain)	What it means in plain language
Linear equations in one variable	Solving for x: one-step, two-step, and multi-step equations
Linear equations in two variables	Equations like $2x + 3y = 12$; intercepts, graphing
Linear functions	Slope, rate of change, $y = mx + b$, interpreting real-world meaning
Systems of two linear equations in two variables	Solving two equations at once (substitution, elimination, graphing)
Linear inequalities in one or two variables	Solving/graphing inequalities, “at least,” “at most,” constraints

Source: College Board SAT Suite, “Algebra” domain overview (satsuite.collegeboard.org).

2. How the Concepts Are Split Across the 7 Sessions

The guiding rule: the first Monday is a full hour of pure teaching, Wednesday of Week 1 is half-teaching / half-practice, and the second Monday still spends 30 minutes teaching the hardest remaining concepts before the program shifts fully into practice and testing. Concepts are sequenced from foundational to hardest, so nothing is taught before its prerequisite.

#	Day	Format	New concepts taught
1	Monday, Wk 1	60 min teaching	Translating words → equations; one/two-step equations; multi-step equations (distributing, variables on both sides); linear inequalities in one variable
2	Wednesday, Wk 1	30 min teach + 30 min practice	Coordinate plane & intercepts; slope (rate of change); slope-intercept form, writing equations of a line
3	Friday, Wk 1	15 min review + Test	Quick recap of Sessions 1–2, then a cumulative quiz on that material
4	Saturday, Wk 1	Office Hours	1:1 remediation on missed test items
5	Monday, Wk 2	30 min teach + 30 min practice	Interpreting linear functions in context; systems of two linear equations (substitution, elimination, special cases) — the two hardest concepts in the unit
6	Wednesday, Wk 2	60 min practice	30 min cumulative mixed practice + 30 min Word Problem Workshop
7	Friday, Wk 2	15 min review + Final Test	Weighted review (heavier on Sessions 5–6), then the cumulative, real College Board–style final assessment (the 70%+ guarantee test) — the last session of the program

Why this order: equations and inequalities (Session 1) are the algebraic “grammar” everything else depends on. Once students can manipulate an equation, Session 2 introduces graphing that same equation. Systems (Session 5) are deliberately taught last and given their own dedicated teaching block, because they require fluency in everything before them. Word problems are held until Session 6, after every underlying tool has been taught.

3. Day-by-Day Lesson Plans

Session 1 — Monday, Week 1 • “Equation & Inequality Bootcamp” (60 min, all teaching)

Objective: by the end of today, students can translate a word phrase into an equation and solve any one-variable linear equation or inequality, including ones with distribution and variables on both sides.

1. Translating words into algebra (10 min)

- “sum,” “more than,” “increased by,” “total” → +
- “difference,” “less than,” “decreased by,” “fewer” → −
- “product,” “times,” “of” → ×
- “quotient,” “per,” “divided by” → ÷

- “is,” “was,” “results in” → =

“Seven less than three times a number is 20.” → $3x - 7 = 20$ → $3x = 27$ → $x = 9$

Watch for: “seven less than three times a number” is $3x - 7$, not $7 - 3x$ — the order flips. This is the single most common SAT wording trap.

2. One- and two-step equations (15 min)

$$x + 15 = -8 \rightarrow x = -23$$

$$4x - 9 = 23 \rightarrow 4x = 32 \rightarrow x = 8$$

$$x/3 + 5 = 2 \rightarrow x/3 = -3 \rightarrow x = -9$$

3. Multi-step equations — distributing & variables on both sides (20 min)

$$3(2x - 4) = 5x + 10 \rightarrow 6x - 12 = 5x + 10 \rightarrow x = 22$$

$$\text{Check: } 3(2 \cdot 22 - 4) = 120; \quad 5(22) + 10 = 120 \quad \checkmark$$

$$(2/3)x + 4 = (1/3)x + 10 \rightarrow (1/3)x = 6 \rightarrow x = 18$$

4. Linear inequalities in one variable (15 min)

$$-2x + 5 > 11 \rightarrow -2x > 6 \rightarrow x < -3 \quad (\div \text{ by a negative flips the sign — the \#1 SAT inequality trap})$$

$$3x - 4 \leq 8 \rightarrow x \leq 4$$

Graphing: open circle = $<$ or $>$; closed circle = $\leq$ or $\geq$.

Home practice (assign for Wednesday): 8 problems — 2 translation, 3 multi-step equations, 3 inequalities. Include the answer key so parents can check alongside their child.

Session 2 — Wednesday, Week 1 • “Graphing & Slope” (30 min teach + 30 min practice)

Objective: by the end of today, students can find intercepts, compute slope from two points, and write a line's equation in slope-intercept form.

5. The coordinate plane & intercepts (10 min)

$$2x + 3y = 12. \quad \text{Set } y=0 \rightarrow x\text{-intercept } (6, 0). \quad \text{Set } x=0 \rightarrow y\text{-intercept } (0, 4).$$

6. Slope — the rate of change (10 min)

$$m = (y_2 - y_1) / (x_2 - x_1). \quad \text{Points } (1,2) \text{ and } (4,11) \rightarrow m = 9/3 = 3$$

Real-world framing (this is how the SAT actually asks it): “A plant is 5 cm tall and grows 3 cm per week” → slope = 3, starting value = 5.

7. Slope-intercept form $y = mx + b$ (10 min)

$$\text{Line through } (2, 5) \text{ with slope } -3: \quad 5 = -3(2) + b \rightarrow b = 11 \rightarrow y = -3x + 11$$

Practice (30 min, individually corrected): 10 mixed problems spanning Session 1 + today's material (spiral review built in from day one), plus 2 real SAT-released questions.

Home practice (due before Session 5): 6 problems on slope/intercepts + 2 review problems from Session 1, to keep equations fresh for Friday's test.

Session 3 — Friday, Week 1 • 15 min Review + Test Day

Review (15 min)

A fast, even-handed recap of Sessions 1–2 — one rapid-fire example of each type (translation, multi-step equation, inequality, intercepts, slope, slope-intercept form) worked on the board, not re-taught from scratch. Spend extra seconds on the two trap questions flagged in Session 1: the “seven less than” wording trap and the “flip the sign” inequality trap — those are the mistakes most likely to resurface under test pressure.

Test

A short, timed, real-College-Board-style quiz (10–12 questions) covering everything from Sessions 1–2: translating expressions, solving equations/inequalities, intercepts, slope, and slope-intercept form. Mix of multiple-choice and grid-in style, matching the Digital SAT format. Score emailed to parents that day.

Session 4 — Saturday, Week 1 • Office Hours

1:1 time to re-teach whatever the Friday test showed was shaky. If most of the group missed the same concept, spend the first 10 minutes re-teaching it to everyone before splitting into individual correction.

Session 5 — Monday, Week 2 • “Interpreting Functions & Systems” (30 min teach + 30 min practice)

This is the hardest content in the unit — treat it as the capstone of the whole program.

8. Interpreting linear functions in context (10 min)

A tutoring company charges $f(x) = 25x + 40$, where x = hours.
40 = flat starting fee (y-intercept); 25 = per-hour rate (slope).
 $f(3) = 25(3) + 40 = 115$

Use function notation ($f(x)$) here even though middle school often uses y — students need to recognize both; this is exactly how the Digital SAT phrases these questions.

9. Systems of two linear equations in two variables (20 min)

- Substitution: $y = 2x+1$ and $3x+y=16 \rightarrow 3x+(2x+1)=16 \rightarrow x=3, y=7 \rightarrow (3, 7)$
- Elimination: $2x+3y=12$ and $4x-3y=6 \rightarrow \text{add} \rightarrow 6x=18 \rightarrow x=3 \rightarrow y=2 \rightarrow (3, 2)$
- Special case — no solution: same slope, different intercept (parallel). E.g., $y=2x+3$ and $y=2x-1$
- Special case — infinite solutions: same slope, same intercept (same line). E.g., $y=2x+3$ and $2y=4x+6$

Practice (30 min, individually corrected): 8–10 problems mixing systems (both methods) and function interpretation. This is the most support-intensive practice block in the program.

Home practice (due before Session 6): 6 systems problems + 2 function-interpretation problems + 2 spiral-review problems.

Session 6 — Wednesday, Week 2 • Full Practice Day (60 min)

Part A — Cumulative mixed practice (30 min): a timed set of 10–12 problems styled after a real Digital SAT math module, covering every concept from Sessions 1–5. Individually corrected in real time.

Part B — Word Problem Workshop (30 min): see Section 4 below.

Session 7 — Friday, Week 2 • 15 min Review + Final Test Day

Review (15 min)

This review is cumulative over the whole program, so don't split it evenly — weight it toward Sessions 5–6 (systems of equations, interpreting linear functions, word problems), since that's the newest, least-practiced material and the most likely source of final-test mistakes. Spend roughly 10 of the 15 minutes there: one quick substitution problem, one elimination problem, one function-interpretation problem, one word problem. Use the remaining 5 minutes for a rapid-fire refresher on the Week 1 basics (an equation, an inequality, a slope question) just to keep them warm — not to re-teach them.

Test

The cumulative, real-College-Board-question assessment referenced in the program's money-back guarantee, and the final session of the program. Every concept from Sessions 1–6 is fair game, weighted toward Sessions 5–6 (systems, interpretation, word problems).

4. Word Problem Workshop (Session 6, Part B)

These six problems are the “hard but fair” tier you asked for — each requires real translation work and at least two steps, but every one is solvable with only the tools taught in Sessions 1–5. Problems 1–3 are Core (every student should finish these); 4–6 are Challenge (stretch problems for students who finish early or want extra practice at home).

Strategy taught first (5 min): (1) circle what you're solving for and assign it a variable, (2) find the “is/equals” moment in the sentence, (3) translate piece by piece using the Session 1 keyword bank, (4) solve, (5) check your answer against the original story, not just the equation.

Core

1. Rate / Distance / Time

Maya leaves her house and bikes toward the library at 12 mph. At the same moment, her brother Leo leaves the library and bikes toward Maya's house at 8 mph. The library is 15 miles from Maya's house. How many minutes after they start will they meet?

Combined closing speed = $12+8 = 20$ mph. $20t = 15 \rightarrow t = 0.75$ hr = 45 minutes

2. Break-Even (linear equations)

A school club is choosing between two fundraiser vendors. Vendor A charges a \$50 setup fee plus \$3 per T-shirt. Vendor B charges a \$20 setup fee plus \$5 per T-shirt. For how many T-shirts do the two vendors cost the same, and what is that cost?

$$50 + 3x = 20 + 5x \rightarrow 30 = 2x \rightarrow x = 15 \text{ shirts; cost} = 50 + 3(15) = \$95$$

3. Mixture

A coffee shop wants to make 20 pounds of a blend selling for \$9/lb by mixing a premium bean (\$12/lb) with a standard bean (\$7/lb). How many pounds of the premium bean are needed?

$$12x + 7(20-x) = 9(20) \rightarrow 12x + 140 - 7x = 180 \rightarrow 5x = 40 \rightarrow x = 8 \text{ lbs premium (12 lbs standard)}$$

Challenge

4. Age Problem

Right now, a father is 3 times as old as his daughter. In 10 years, he will be twice as old as she will be then. How old is the daughter now?

$$3d + 10 = 2(d+10) \rightarrow 3d+10 = 2d+20 \rightarrow d = 10 \text{ (father is 30). Check: } 40 = 2(20) \checkmark$$

5. Budget Constraint (inequality)

Jordan has \$60 for movie tickets and popcorn. Each ticket costs \$9.50, and a large popcorn costs \$6. If Jordan buys 4 tickets, what is the maximum number of popcorns he can buy without going over budget?

$$4(9.50) = 38. \text{ Remaining} = 60 - 38 = 22. \quad 6p \leq 22 \rightarrow p \leq 3.67 \rightarrow 3 \text{ popcorns (round down)}$$

6. Linear Function in Context (Digital SAT style)

A candle is 20 cm tall when lit and burns at a constant rate. After burning for 5 hours, it is 12.5 cm tall. Write a function $h(t)$ for its height after t hours, and find how many hours until it burns out completely.

$$\text{slope} = (12.5 - 20)/5 = -1.5 \text{ cm/hr. } h(t) = -1.5t + 20. \quad h(t) = 0 \rightarrow t = 40/3 \approx 13.3 \text{ hr} = 13 \text{ hr } 20 \text{ min}$$

Take-home extension: 3 additional word problems (one of each Core type, new numbers) for students who want more reps before the Friday final.

5. Parent-Facing At-a-Glance Schedule

Share this table directly with parents — it answers “what is my child learning today?” at a glance, per the program’s transparency commitment.

Day	Format	What your child is learning
Mon, Wk 1	Teaching (60 min)	Turning word problems into equations; solving equations and inequalities for x
Wed, Wk 1	Teach (30) + Practice (30)	Graphing lines, slope as “rate of change,” writing a line's equation
Fri, Wk 1	Review (15) + Test Day	Quick recap, then a quiz covering everything from Monday & Wednesday
Sat, Wk 1	Office Hours	1:1 help on anything missed Friday
Mon, Wk 2	Teach (30) + Practice (30)	What slope/intercept mean in real situations; solving two equations at once (systems)
Wed, Wk 2	Practice (30) + Word Problems (30)	Cumulative review + real-world challenge word problems
Fri, Wk 2	Review (15) + Final Test	Weighted review (systems, functions, word problems), then the cumulative final assessment (counts toward the 70%+ guarantee) — last session of the program

Fill in actual calendar dates and Zoom times once scheduled — this table is designed to be dropped into an enrollment email as-is.

6. Materials & At-Home Support

Materials needed each session

Pencil, notebook or graph paper (for Sessions 2, 5, 6), a basic or scientific calculator (not required, but allowed per the placement exam's policy), device for Zoom.

How parents can help between sessions

You don't need to know the math — just ask your child to explain their homework answer out loud before checking it against the answer key. If they can explain why an equation is set up the way it is, they've actually learned the concept, not just matched a pattern.

Difficulty calibration note

Every practice and homework set is built in three informal tiers — Warm-up (confirms the basic mechanic), Core (SAT-typical difficulty), and Challenge (the word-problem tier above). No student sees only Warm-up or only Challenge; everyone works through all three so pacing stays consistent with the rest of the cohort while still stretching stronger students.